



HCI and Computer Graphics Assignment

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Task1: Display Pixel Density (PPI/DPI) Calculator

Calculated DPI values for a desktop

```
Enter horizontal resolution(pixels): 1920
Enter vertical resolution(pixels): 1080
Enter physical diagonal size(inches): 24

--- DISPLAY MATRICES ANALYSIS---
Total pixel count : 2,073,600 pixels
Aspect Ratio : 16:9
Calculated DPI : 91.79 DPI
Denisty Category : Low Density (Standard Monitor)
```

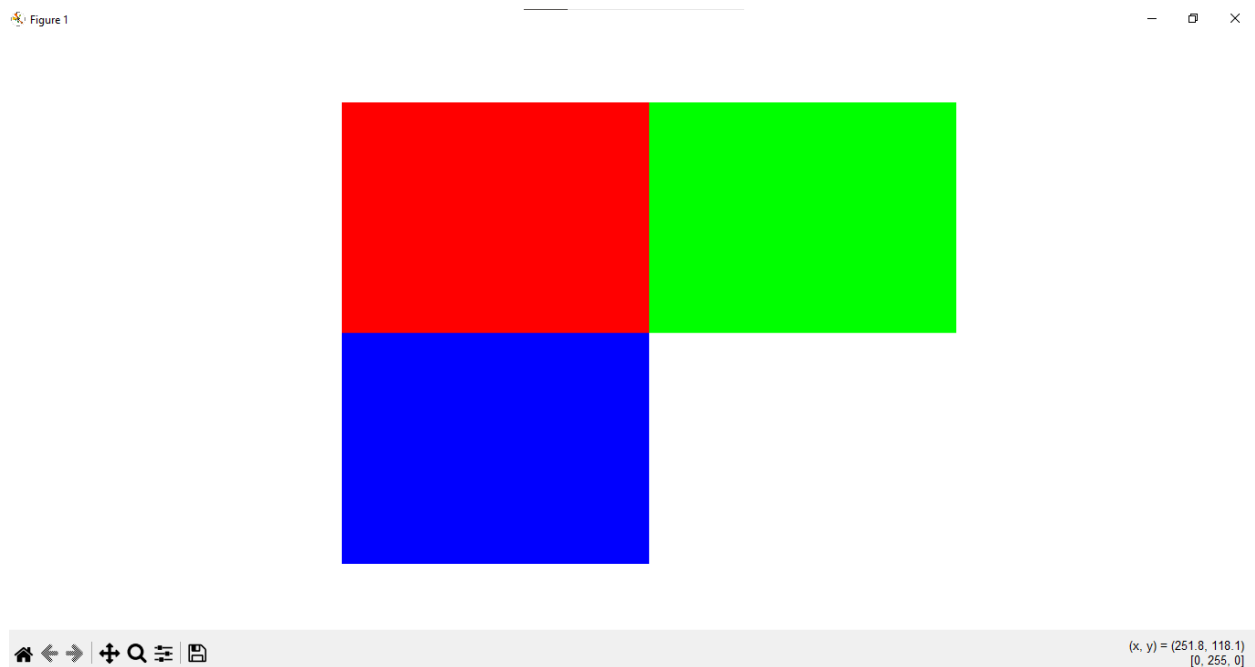
Calculated DPI values for a smartphone screen:

```
Enter horizontal resolution(pixels): 1170
Enter vertical resolution(pixels): 2532
Enter physical diagonal size(inches): 6.1

--- DISPLAY MATRICES ANALYSIS---
Total pixel count : 2,962,440 pixels
Aspect Ratio : 195:422
Calculated DPI : 457.25 DPI
Denisty Category : High Density (Retina / Mobile)
```

Task2: Environment Setup & Synthetic Image Matrix Creation

Figure:



Output:

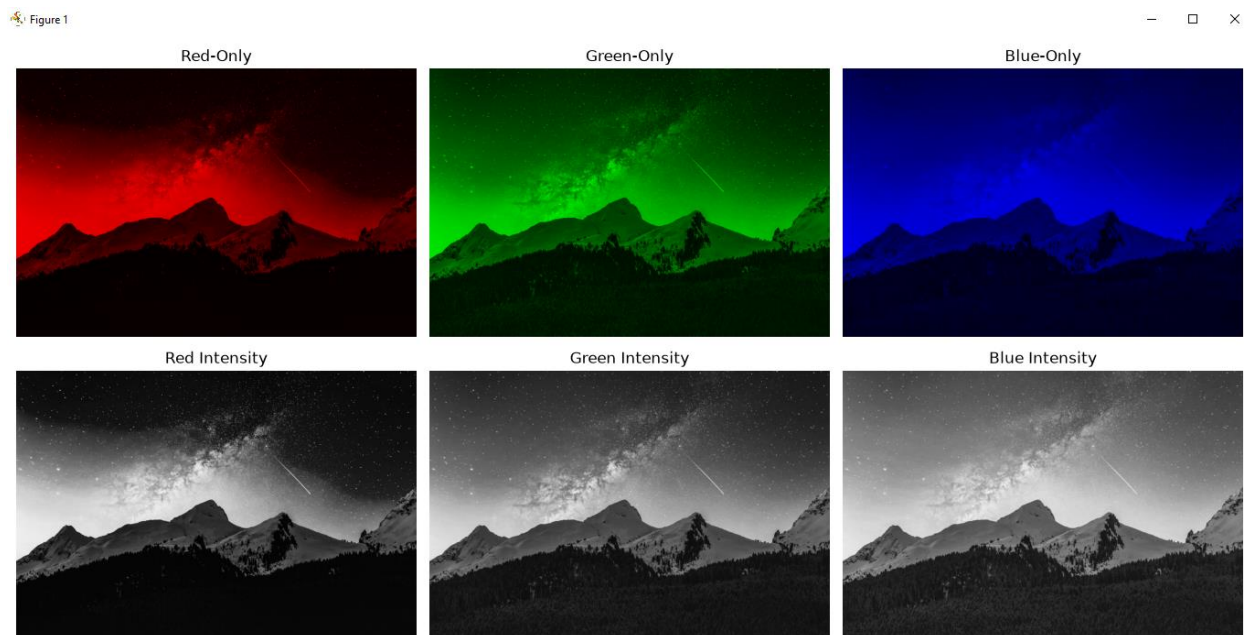
```
--- SYNTHETIC MATRIX METRICS ---  
Array Shape (H, W, C) : (300, 400, 3)  
Data Type : uint8  
Total Elements : 360,000 values  
Memory Footprint : 360,000 bytes (351.56 KB)
```

Task3: Channel Slicing & Isolation

Sample Image:



Figure:



Output:

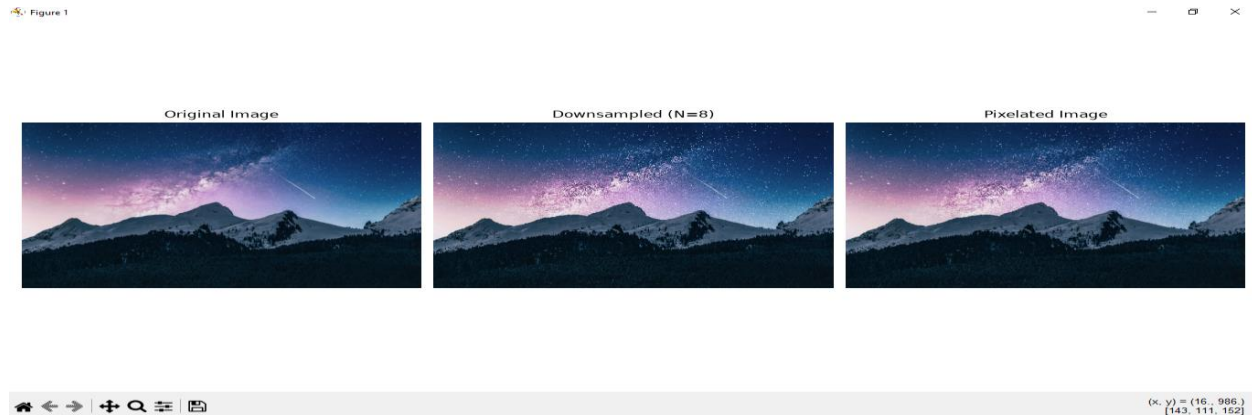
```
--- CHANNEL EXTRACTION SUMMARY ---  
Original Image Shape : (2733, 4096, 3)  
Red Channel 2D Shape : (2733, 4096) | Mean Intensity: 55.47  
Green Channel 2D Shape : (2733, 4096) | Mean Intensity: 71.25  
Blue Channel 2D Shape : (2733, 4096) | Mean Intensity: 100.00  
Display Window : Matplotlib 2x3 Subplot Grid Rendered.
```

Task4: Spatial Downsampling & Pixelation via Striding

Sample Image:



Figure:



Output:

```
--- DOWNSAMPLING ANALYSIS (N = 8) ---  
Original Shape : (2733, 4096, 3) | Memory: 33,583,104 bytes  
Downsampled Shape : (342, 512, 3) | Memory: 525,312 bytes  
Re-expanded Shape : (2733, 4096, 3) | Visual: Blocky Pixelation  
Dimension Reduction: 98.44% reduction per axis  
Memory Savings : 98.44% data reduction
```

Why does adding an Alpha channel (RGBA) increase an image array's memory consumption by 33% compared to standard RGB?

RGBA uses 4 channels (R, G, B, Alpha), while RGB uses only 3 channels. Since each uint8 channel takes 1 byte, adding Alpha adds 1 extra byte per pixel.

So, RGBA uses 4 bytes instead of 3, which means memory increases by about 33%.

